

§ 111. The resolvents of the icosahedral equation.

The formulation of the problem which we gave at the end of the preceding paragraph leads us to the question of the different resolvents of the icosahedral equation. To every divisor of the icosahedral group there corresponds such a resolvent, whose degree is equal to the index of the divisor, and therefore is always a divisor of 60. The icosahedral group contains:

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|-----------------------------------|-------------------------|
| 1) tetrahedral groups (§ 56) : | resolvents of degree 5, |
| 2) dihedral groups D_5 (§ 55) : | of degree 6, |
| 3) dihedral groups D_3 : | of degree 10, |
| 4) cyclic groups C_5 : | of degree 12, |
| 5) four-groups D_2 : | of degree 15, |
| 6) cyclic groups C_3 : | of degree 20, |
| 7) cyclic groups C_2 : | of degree 30. |

Since the icosahedral group is simple, all these equations are total resolvents of one another; here we restrict ourselves to the consideration of the most important among them.

The resolvents of the lowest degree, namely the fifth, are those belonging to the tetrahedral group. To find them we must seek a function of ϑ that remains unchanged under the substitutions of the tetrahedral group, while it is five-valued in the icosahedral group.

We take the icosahedral group in the normal form set up in § 58, (22), and select from it the tetrahedral group Q (§ 59), constructed over the four-group

$$1, \quad \psi, \quad \chi, \quad \psi\chi, \quad (1)$$

where, if ε is an imaginary fifth root of unity, ψ and χ have the meanings

$$\psi(x) = -\frac{1}{x}, \quad \chi(x) = \frac{\omega x + 1}{x - \omega}, \quad \omega = \varepsilon + \varepsilon^{-1}. \quad (2)$$

If we put $\Theta(x) = \varepsilon x$, then the whole icosahedral group is

$$P = Q + Q\Theta + Q\Theta^2 + Q\Theta^3 + Q\Theta^4, \quad (3)$$

and

$$Q_h = \Theta^{-h} Q \Theta^h \quad (4)$$

are the tetrahedral groups conjugate to Q .

In § 56 we learned two forms f and H of the sixth and eighth degrees which are left absolutely unchanged by the tetrahedral substitutions when these are taken homogeneously and with determinant 1. These forms cannot be used at once in the form given there, because the tetrahedral group was there assumed in another shape. But the first of these forms is found very simply if we recall that its roots were the double poles of the tetrahedral group, hence the roots of the three equations

$$x = \psi(x), \quad x = \chi(x), \quad x = \psi\chi(x),$$

or

$$x^2 + 1 = 0, \quad x^2 - 2\omega x - 1 = 0, \quad \omega x^2 + 2x - \omega = 0.$$

Using the relation

$$\omega^2 + \omega = 1$$

one obtains the equation for the double poles

$$(x^2 + 1)(x^4 + 2x^3 - 6x^2 - 2x + 1) = 0.$$

Thus the required tetrahedral form of degree 6 in the homogeneous variables x_1, x_2 is

$$t(x_1, x_2) = x_1^6 + 2x_1^5x_2 - 5x_1^4x_2^2 - 5x_1^2x_2^4 - 2x_1x_2^5 + x_2^6. \quad (5)$$

The second tetrahedral form sought is the Hessian determinant of this one. If we set

$$\tilde{\omega}(x_1, x_2) = \frac{1}{5^2 \cdot 4^2} \{t''(x_1, x_1)t''(x_2, x_2) - [t''(x_1, x_2)]^2\},$$

then a simple calculation gives

$$\begin{aligned} \tilde{\omega}(x_1, x_2) = & -(x_1^8 + x_2^8) + (x_1^7 x_2 - x_1 x_2^7) - 7(x_1^6 x_2^2 + x_1^2 x_2^6) \\ & - 7(x_1^5 x_2^3 - x_1^3 x_2^5). \end{aligned} \quad (6)$$

We further record the three icosahedral forms of degrees 12, 20, and 30, which likewise remain unchanged under the tetrahedral substitutions (§60):

$$f(x_1, x_2) = x_1 x_2 (x_1^{10} + 11x_1^5 x_2^5 - x_2^{10}), \quad (7)$$

$$H(x_1, x_2) = -(x_1^{20} + x_2^{20}) + 228(x_1^{15} x_2^5 - x_1^5 x_2^{15}) - 494x_1^{10} x_2^{10}, \quad (8)$$

$$\begin{aligned} T(x_1, x_2) = & (x_1^{30} + x_2^{30}) + 522(x_1^{25} x_2^5 - x_1^5 x_2^{25}) \\ & - 10005(x_1^{20} x_2^{10} + x_1^{10} x_2^{20}). \end{aligned} \quad (9)$$

If from these invariant forms of the tetrahedron we form functions of $x_1 : x_2$ and put $\vartheta = x_1 : x_2$, the result is functions left unchanged by the linear fractional tetrahedral substitutions, and hence roots of resolvents of degree 5 of the icosahedral equation. The icosahedral equation itself is obtained in two forms if we put

$$H^3 = z f^5, \quad T^2 = z_1 f^5, \quad z + z_1 = 1728. \quad (10)$$

The resolvents depend rationally on z .

For the moment let $\varphi(x_1, x_2)$ denote any absolutely invariant form of the tetrahedron. Under any one of the substitutions $Q\Theta$ it passes into

$$\varphi_\varepsilon = \varphi(\varepsilon^{-2}x_1, \varepsilon^2x_2), \quad (11)$$

and this function remains unchanged under the substitutions of the group $\Theta^{-1}Q\Theta$. It follows that every symmetric function of the five forms φ_ε is an invariant of the icosahedral group, and hence, by the theorem of §61, can be expressed by the ground forms f, H, T . This theorem permits a comparatively easy calculation of the resolvents.

The simplest functions which we can use as roots of resolvents are

$$r = \frac{t^2}{f}, \quad u = \frac{f^2 t}{T}, \quad v = \frac{f \tilde{\omega}}{H}. \quad (12)$$

We first observe that the symmetric ground functions of the five functions t_ε are icosahedral invariants of degrees 6, 12, 18, 30. Since, by the theorems in §61, the degrees 6 and 18 do not occur among these invariants, an equation of the form

$$t^5 + a f t^3 + b f^2 t + c T = 0 \quad (13)$$

follows, where a, b, c are numerical coefficients. To determine them, let $S(\varphi)$ denote the sum of the five functions φ_ε in (11), and let $\Pi(\varphi)$ denote their product. Newton's formulae (Vol. I, §42) give

$$a f = -\frac{1}{2}S(t^2), \quad b f^2 = \frac{1}{2}a^2 f^2 - \frac{1}{4}S(t^4), \quad c T = -\Pi(t). \quad (14)$$

In the sums $S(\varphi)$, only those terms $x_1^h x_2^k$ have a non-zero coefficient for which $h - k$ is divisible by 5. Hence a, b, c are obtained by equating the coefficients of $x_1^{11} x_2, x_1^{22} x_2^2, x_1^{30}$ on both sides of (14), and one finds the relation

$$t^5 - 10 f t^3 + 45 f^2 t - T = 0, \quad (15)$$

or also

$$t^2(t^4 - 10ft^2 + 45f^2)^2 - T^2 = 0. \quad (16)$$

From (16) and (15), using (10) and (12), the equations of degree 5 for r and for u are

$$r(r^2 - 10r + 45)^2 = z_1, \quad (17)$$

$$u^5 - \frac{10u^3}{z_1} + \frac{45u}{z_1^2} - \frac{1}{z_1^2} = 0. \quad (18)$$

The latter of these equations is transformed by the substitution

$$y = -\frac{1}{3u}, \quad z_1 = 27\gamma \quad (19)$$

into the form

$$y^5 + 15y^4 - 10\gamma y^2 + 3\gamma^2 = 0, \quad (20)$$

which in § 75 of the first volume we already learned as a normal form of the equation of degree 5.

§ 112. The principal resolvent of degree five.

If one uses the functions u, v simultaneously, one can derive a family of resolvents having the form of the principal equation of degree 5, and one can bring it directly into agreement with an arbitrarily given principal equation of degree 5.¹

Since for the icosahedron there are no invariants of degrees 8, 14, 16, 22, or 28, the functions $S(\tilde{\omega}), S(t\tilde{\omega}), S(\tilde{\omega}^2), S(t\tilde{\omega}^2), S(t^2\tilde{\omega}^2)$ must vanish. Thus, if we put

$$Y = \alpha\tilde{\omega} + \beta t\tilde{\omega}, \quad (1)$$

the functions

$$S(Y), \quad S(Y^2)$$

also vanish for arbitrary values of the parameters α, β . From this there results an identity of the form

$$Y^5 + 5aY^2 + 5bY + c = 0, \quad (2)$$

where a, b, c denote homogeneous functions of degrees 3, 4, and 5 in α, β , whose coefficients are icosahedral invariants.

The function c is easily determined from the formula

$$c = -\Pi(\tilde{\omega})\Pi(\alpha + \beta t).$$

Indeed, by (15) of the preceding paragraph, for an indeterminate λ ,

$$\lambda^5 - 10f\lambda^3 + 45f^2\lambda - T = \Pi(\lambda - t),$$

and hence, setting $\lambda = -\alpha : \beta$,

$$\Pi(\alpha + \beta t) = \alpha^5 - 10f\alpha^3\beta^2 + 45f^2\alpha\beta^4 + T\beta^5.$$

Moreover, by comparison of one term [§ 111, (6), (8)] one obtains immediately

$$\Pi(\tilde{\omega}) = -H^2,$$

and therefore

$$c = H^2(\alpha^5 - 10f\alpha^3\beta^2 + 45f^2\alpha\beta^4 + T\beta^5). \quad (3)$$

¹Kiepert, *Göttinger Nachrichten* 1878. Crelle's Journal, vol. 87. Klein, *Ikosäeder*, p. 106.

The coefficients a, b are most simply calculated with the help of Newton's formulae (Vol. I, § 42):

$$-15a = S(\alpha\tilde{\omega} + \beta t\tilde{\omega})^3, \quad -20b = S(\alpha\tilde{\omega} + \beta t\tilde{\omega})^4.$$

One uses the formulae which, after a short calculation, are obtained by comparing a single term on the right and left sides:

$$\begin{aligned} S(\tilde{\omega}^3) &= -3 \cdot 5 \cdot 8f^2, & S(t\tilde{\omega}^3) &= -5T, \\ S(t^2\tilde{\omega}^3) &= -5 \cdot 72f^3, & S(t^3\tilde{\omega}^3) &= -3 \cdot 5fT, \\ S(\tilde{\omega}^4) &= 4 \cdot 5fH, & S(t^2\tilde{\omega}^4) &= -5 \cdot 12f^2H, \\ S(t^3\tilde{\omega}^4) &= -5HT, & S(t^4\tilde{\omega}^4) &= -4 \cdot 5 \cdot 27f^3H. \end{aligned}$$

Thus one finds

$$a = 8f^2\alpha^3 + T\alpha^2\beta + 72f^3\alpha\beta^2 + fT\beta^3, \quad (4)$$

$$b = -fH\alpha^4 + 18f^2H\alpha^2\beta^2 + HT\alpha\beta^3 + 27f^3H\beta^4. \quad (5)$$

To obtain from this the resolvent of the icosahedral equation, we must introduce the functions u, v [§ 111, (12)] instead of $\tilde{\omega}$ and t .

If, denoting by λ, μ two arbitrary undetermined quantities, we set

$$\alpha = \frac{\lambda f}{H}, \quad \beta = \frac{\mu f^3}{HT}, \quad (6)$$

then from (1) and § 111, (12) there follows

$$Y = \lambda v + \mu v u. \quad (7)$$

And if, as in § 111, we put

$$\frac{H^3}{f^5} = z, \quad \frac{T^2}{f^5} = z_1, \quad z + z_1 = 1728, \quad (8)$$

then the expressions (3), (4), (5) pass into the following:

$$\begin{aligned} az &= 8\lambda^3 + \lambda^2\mu + \frac{72\lambda\mu^2 + \mu^3}{z_1}, \\ bz &= -\lambda^4 + \frac{18\lambda^2\mu^2 + \lambda\mu^3}{z_1} + 27\frac{\mu^4}{z_1^2}, \\ cz &= \lambda^5 - 10\frac{\lambda^3\mu^2}{z_1} + \frac{45\lambda\mu^4 + \mu^5}{z_1^2}, \end{aligned} \quad (9)$$

where Y is a root of the equation

$$Y^5 + 5aY^2 + 5bY + c = 0. \quad (10)$$

This equation has the form of a principal equation of degree 5, and it can represent every principal equation if we may assume a, b, c arbitrarily. Thus, if a, b, c are assumed given arbitrarily, the three quantities λ, μ, z_1 and $z = 1728 - z_1$ have to be determined from (9). A very remarkable property of this system of equations is that these unknowns can be expressed rationally by the given quantities a, b, c and the discriminant of equation (10).

To prove this, we first derive simpler equations from (9). If the second of them is multiplied by λ and added to the third, then

$$z_1(\lambda b + c) = \mu^2 a \quad (11)$$

follows. If the third is multiplied by λ , and the second by $\mu^2 : z_1$, and the latter is subtracted, one obtains

$$z \left(\lambda c - \frac{\mu^2}{z_1} b \right) = \left(\lambda^2 - 3 \frac{\mu^2}{z_1} \right)^3. \quad (12)$$

In the same way the first and second give

$$z \frac{a\lambda + 8b}{\mu} = \lambda^3 + 216 \frac{\lambda^2 \mu}{z_1} + 9 \frac{\lambda \mu^2}{z_1} + 216 \frac{\mu^3}{z_1^2}.$$

Squaring both sides of this equation gives

$$\begin{aligned} z_1^2 z^2 \left(\frac{a\lambda + 8b}{\mu} \right)^2 &= \lambda^6 z_1 + 2 \cdot 216 \lambda^5 \mu \\ &\quad + \left(18 + \frac{216^2}{z_1} \right) \lambda^4 \mu^2 + \frac{2 \cdot 2160}{z_1} \lambda^3 \mu^3 \\ &\quad + \left(\frac{81}{z_1} + \frac{2 \cdot 216^2}{z_1^2} \right) \lambda^2 \mu^4 + \frac{18 \cdot 216}{z_1^2} \lambda \mu^5 + \frac{216^2 \mu^6}{z_1^3}. \end{aligned}$$

Subtracting this from the formula derived from the first equation (9),

$$\begin{aligned} 27a^2 z^2 &= 1728 \lambda^6 + 2 \cdot 216 \lambda^5 \mu + 27 \left(1 + \frac{16 \cdot 72}{z_1} \right) \lambda^4 \mu^2 \\ &\quad + \frac{2 \cdot 2160}{z_1} \lambda^3 \mu^3 + 27 \left(\frac{2}{z_1} + \frac{72^2}{z_1^2} \right) \lambda^2 \mu^4 + \frac{18 \cdot 216}{z_1^2} \lambda \mu^5 + \frac{27 \mu^6}{z_1^2}, \end{aligned}$$

then, using (8), one obtains

$$z \left[27a^2 - z_1 \left(\frac{a\lambda + 8b}{\mu} \right)^2 \right] = \left(\lambda^2 - 3 \frac{\mu^2}{z_1} \right)^3. \quad (13)$$

Further comparison with (12) yields

$$27a^2 - \frac{z_1}{\mu^2} (a\lambda + 8b)^2 = \lambda c - \frac{\mu^2}{z_1} b. \quad (14)$$

Eliminating from this, by means of (11), the ratio $\mu^2 : z_1$, gives the quadratic equation for λ :

$$\begin{aligned} \lambda^2 (a^4 + abc - b^3) - \lambda (11a^3b - ac^2 + 2b^2c) \\ - 27a^3c + 64a^2b^2 - bc^2 = 0. \end{aligned} \quad (15)$$

The discriminant of this quadratic equation is

$$\begin{aligned} \Delta &= (11a^3b - ac^2 + 2b^2c)^2 \\ &\quad + 4(a^4 + abc - b^3)(27a^3c - 64a^2b^2 + bc^2) \\ &= a^2(108a^5c - 135a^4b^2 + 90a^2bc^2 - 320ab^3c + 256b^5 + c^4). \end{aligned}$$

In § 74 of the first volume the discriminant D of the principal equation of degree 5 was derived. If in the formula (3) found there one puts

$$a_0 = 1, \quad a_3 = 5a, \quad a_4 = 5b, \quad a_5 = c,$$

then

$$D = 5^5(108a^5c - 135a^4b^2 + 90a^2bc^2 - 320ab^3c + 256b^5 + c^4),$$

and hence

$$5^5 \Delta = a^2 D. \quad (16)$$

From (15) one therefore obtains for λ the expression

$$\lambda = \frac{11a^3b - ac^2 + 2b^2c \pm \frac{1}{25}a\sqrt{\frac{1}{5}D}}{2(a^4 + abc - b^3)},$$

in which the square root may have either sign. Thus, besides the square root of the discriminant, which is rationally expressible by the roots, $\sqrt{5}$ also occurs.

Once λ has been calculated, (11) gives the ratio $z_1 : \mu^2$, and (12) gives the last unknown z rationally in terms of λ .

§ 113. Resolvents of the sixth degree.

To find the resolvents of degree 6 of the icosahedral equation, we start from one of the dihedral groups D_5 contained in the icosahedral group (§ 111, 2). The representation of the icosahedral group [§ 58, (20)] gives immediately the decomposition into cosets. If

$$Q = \Theta^r, \quad \Theta^r \psi, \quad r = 0, 1, 2, 3, 4 \quad (1)$$

is the dihedral group from which we start, then we obtain the full icosahedral group as

$$P = Q + Q\chi + Q\chi\Theta + Q\chi\Theta^2 + Q\chi\Theta^3 + Q\chi\Theta^4. \quad (2)$$

If U_∞ is a function left unchanged by the substitutions of Q , and is transformed by

$$\chi, \quad \chi\Theta, \quad \chi\Theta^2, \quad \chi\Theta^3, \quad \chi\Theta^4$$

into

$$U_0, \quad U_1, \quad U_2, \quad U_3, \quad U_4,$$

then $U_\infty, U_0, U_1, U_2, U_3, U_4$ are the roots of a resolvent of degree 6. Concerning the notation U_∞ , compare § 65.

To obtain the Galois group of this equation of degree 6, we must investigate the influence exerted by the application of the icosahedral substitutions upon the system of cosets (2). It suffices to consider the substitutions Θ, ψ, χ . But, observing the order of the cosets in (2), by § 58, (23), (25),

$$\begin{aligned} P\Theta &= Q + Q\chi\Theta + Q\chi\Theta^2 + Q\chi\Theta^3 + Q\chi\Theta^4 + Q\chi, \\ P\psi &= Q + Q\chi + Q\chi\Theta^4 + Q\chi\Theta^3 + Q\chi\Theta^2 + Q\chi\Theta, \\ P\chi &= Q\chi + Q + Q\chi\Theta + Q\chi\Theta^3 + Q\chi\Theta^2 + Q\chi\Theta^4. \end{aligned}$$

It follows that the indices of U are permuted in the following way:

	∞	0	1	2	3	4
$\Theta)$	∞	1	2	3	4	0
$\psi)$	∞	0	4	3	2	1
$\chi)$	0	∞	1	3	2	4

Denoting the index generally by ξ , and taking ξ , as in the seventh section, modulo 5, so that two numbers congruent modulo 5 are not regarded as different, we may write these permutations in the form

$$\Theta = (\xi, \xi + 1), \quad \psi = (\xi, -\xi), \quad \chi = \left(\xi, \frac{1}{\xi} \right).$$

It follows that the group of our equation of degree 6 agrees with the congruence group L_5 (§ 68).

In order to arrive at the formation of resolvents of degree 6, we must proceed in a way similar to that used for the resolvents of degree 5.

We have already learned in § 55 the ground forms of the dihedral groups. We now write them in the form

$$\varphi_1 = x_1 x_2, \quad \varphi_2 = x_1^5 + i x_2^5, \quad \varphi_3 = x_1^5 - i x_2^5.$$

Taking Θ and ψ with determinant 1, thus

$$\Theta = \begin{pmatrix} e^{\pi i/5} & 0 \\ 0 & e^{-\pi i/5} \end{pmatrix}, \quad \psi = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

the ground forms $\varphi_1, \varphi_2, \varphi_3$ change as follows:

	φ_1	φ_2	φ_3
Θ)	φ_1	$-\varphi_2$	$-\varphi_3$
ψ)	$-\varphi_1$	$-i\varphi_2$	$i\varphi_3$

Therefore $\varphi_1^2, \varphi_2^4, \varphi_3^4, \varphi_2 \varphi_3$ are absolutely invariant. Hence, if as in § 111 we understand by f, H, T the icosahedral invariants, then the following functions may be introduced as roots of resolvents of degree 6:

$$\frac{\varphi_1^2 H}{f^2}, \quad \frac{\varphi_2^4}{H}, \quad \frac{\varphi_3^4}{H}, \quad \frac{\varphi_2 \varphi_3 H}{T}.$$

To obtain the expressions for the remaining roots, one must investigate the effect of the substitutions $\Theta^r \chi$, taken with determinant 1, on the functions $\varphi_1, \varphi_2, \varphi_3$. We shall carry this out only for the function φ_1 , which leads to the simplest resolvent of degree 6.

If in φ_1 we apply the substitution χ in the form § 58, (26), that is, replace x_1, x_2 by

$$\frac{x_1}{\varepsilon - \varepsilon^{-1}} + \frac{x_2}{\varepsilon^2 - \varepsilon^{-2}}, \quad \frac{x_1}{\varepsilon^2 - \varepsilon^{-2}} - \frac{x_2}{\varepsilon - \varepsilon^{-1}},$$

then φ_1 passes into

$$\frac{-x_1^2 + i x_1 x_2 + x_2^2}{\sqrt{5}}.$$

Accordingly we set

$$w_\infty = 5\varphi_1^2 = 5x_1^2 x_2^2, \tag{3}$$

and by applying the substitutions $\chi \Theta^r$ obtain from it the five other forms

$$w_r = (\varepsilon^r x_1^2 + x_1 x_2 - \varepsilon^{-r} x_2^2)^2. \tag{4}$$

The symmetric functions of the six forms w are now invariant icosahedral forms, and therefore one can easily form the coefficients of the equation whose roots are the w .

By comparing the degrees and one coefficient in each case for the power sums of the w , we obtain

$$S(w) = 0, \quad S(w^2) = 0, \quad S(w^3) = 30f, \quad S(w^4) = 0, \quad S(w^5) = -5H,$$

and thus the product of all six w is

$$\Pi(w) = 5f^2.$$

Hence Newton's formulae (Vol. I, § 42) give, for w , the equation

$$w^6 - 10fw^3 + Hw + 5f^2 = 0. \tag{5}$$

Putting therefore

$$U = \frac{wH}{f^2}, \quad z = \frac{H^3}{f^2}, \tag{6}$$

we obtain from it the equation of degree 6 for U :

$$U^6 - 10zU^3 + z^2U + 5z^2 = 0, \quad (7)$$

as a resolvent of degree 6 of the icosahedral equation.

If, according to (3) and (4), we put

$$\sqrt{w_\infty} = \sqrt{5}x_1x_2, \quad \sqrt{w_r} = \varepsilon^r x_1^2 + x_1x_2 - \varepsilon^{-r}x_2^2, \quad (8)$$

then the following relations result:

$$\begin{aligned} \sqrt{5}\sqrt{w_\infty} &= \sqrt{w_0} + \sqrt{w_1} + \sqrt{w_2} + \sqrt{w_3} + \sqrt{w_4}, \\ 0 &= \sqrt{w_0} + \varepsilon^2\sqrt{w_1} + \varepsilon^4\sqrt{w_2} + \varepsilon\sqrt{w_3} + \varepsilon^3\sqrt{w_4}, \\ 0 &= \sqrt{w_0} + \varepsilon^3\sqrt{w_1} + \varepsilon\sqrt{w_2} + \varepsilon^4\sqrt{w_3} + \varepsilon^2\sqrt{w_4}, \end{aligned} \quad (9)$$

and

$$\frac{x_1\sqrt{5}}{x_2} = \frac{\sqrt{w_0} + \varepsilon^4\sqrt{w_1} + \varepsilon^3\sqrt{w_2} + \varepsilon^2\sqrt{w_3} + \varepsilon\sqrt{w_4}}{\sqrt{w_\infty}}. \quad (10)$$

Since these equations are homogeneous, the w in (9), (10) may also be replaced by the U .

Jacobi first pointed out the equations of degree 6 whose roots satisfy the relations (9); they are therefore called Jacobi equations of degree 6.²

§ 114. On the solution of the icosahedral equation by transcendental functions.

Here, without entering into proofs or more detailed developments, we shall briefly indicate the relations which exist between the investigations carried out above and the theory of certain transcendental functions, especially the theory of elliptic functions and the theory of hypergeometric series, from which the solution of the icosahedral equation, and with it that of all its resolvents, by transcendental functions results. This leads out of the domain of algebra, but it stands to it in about the same relation as the use of logarithms in the computation of roots, or the use of trisection of angles in cubic equations.

We must here restrict ourselves to a brief statement of the results and will make no applications of them in the sequel, so that these developments, which can be fully intelligible only to a reader somewhat familiar with the theory of transcendental functions, may be passed over without disturbing the connection.

Let ω denote a variable whose imaginary part is positive, so that

$$q = e^{\pi i \omega} \quad (1)$$

is a quantity whose absolute value is a proper fraction. We shall consider the following three functions of this variable ω :

$$\begin{aligned} \eta(\omega) &= q^{1/12} \prod_{v=1}^{\infty} (1 - q^{2v}) \\ &= q^{1/12} \sum_{v=-\infty}^{\infty} (-1)^v q^{3v^2+v} = \sum_{v=-\infty}^{\infty} (-1)^v q^{(6v+1)^2/12}, \end{aligned} \quad (2)$$

$$f(\omega) = q^{-1/24} \prod_{v=1}^{\infty} (1 + q^{2v-1}), \quad (3)$$

$$\gamma_2(\omega) = \frac{f(\omega)^{24} - 16}{f(\omega)^8}. \quad (4)$$

²*Suite des notices sur les fonctions elliptiques.* Crelle's Journal, vol. III (1828); Works, vol. I, p. 261.

In the theory of transformations of elliptic functions³ an equation of degree 6 is derived whose roots are the six quantities

$$v_{\infty} = 5 \left(\frac{\eta(5\omega)}{\eta(\omega)} \right)^2, \quad v_{\xi} = \left(\frac{\eta((-24\xi + \omega)/5)}{\eta(\omega)} \right)^2, \quad (5)$$

where ξ runs through a complete system of residues modulo 5. This equation is

$$v^6 + 10v^3 - \gamma_2(\omega)v + 5 = 0, \quad (6)$$

and, as one sees, has a close resemblance to the resolvent of degree 6 considered in the preceding paragraph,

$$U^6 - 10zU^3 + z^2U + 5z^2 = 0. \quad (7)$$

The two equations pass directly into each other if one puts

$$z = -\gamma_2(\omega)^3, \quad (8)$$

$$U = -\gamma_2(\omega)v. \quad (9)$$

The equation (6) has the same Galois group as (7).

We may now put any one of the roots of (7) equal to U_{∞} , and consequently we may let U_{∞} and v_{∞} correspond to one another in (9). Then we may take any one of the other roots of (7) for U_0 , since in § 113, (4) we may replace x_1, x_2 by $\varepsilon^a x_1, \varepsilon^{-a} x_2$. Thus we let U_0 and v_0 correspond. Finally, we may choose the imaginary fifth root of unity ε in such a way that U_1 and v_1 correspond. Then it follows from the agreement of the group that in general U_{ξ} and v_{ξ} correspond, so that

$$U_{\xi} = -\gamma_2(\omega)v_{\xi}, \quad \xi = \infty, 0, 1, 2, 3, 4. \quad (10)$$

Thus, if for a given z we determine ω from the transcendental equation (8), then (10) and (6) give the complete solution of the icosahedral resolvent of degree 6.

By (2) and (5), for $\xi = 0, 1, 2, 3, 4$,

$$\eta(\omega)\sqrt{v_{\xi}} = \sum_{-\infty}^{\infty} (-1)^v e^{\frac{\pi i}{60}(\omega - 24\xi)(6v+1)^2}. \quad (11)$$

The exponent $(6v+1)^2$ can, modulo 5, give only the residue 0, 1, -1. If therefore v_1, v_2 run through all the numbers satisfying

$$v_1 \equiv 1, 2, \quad v_2 \equiv 0, 3 \pmod{5},$$

then the expression (11) can be decomposed in the form

$$\begin{aligned} \eta(\omega)\sqrt{v_{\xi}} &= \eta(5\omega) + \varepsilon^{\xi} \sum_{v_1} (-1)^{v_1} e^{\frac{\pi i}{60}\omega(6v_1+1)^2} \\ &\quad + \varepsilon^{-\xi} \sum_{v_2} (-1)^{v_2} e^{\frac{\pi i}{60}\omega(6v_2+1)^2}, \end{aligned} \quad (12)$$

where $\varepsilon = e^{2\pi i/5}$. From this one recognizes that the quantities $\sqrt{v_{\xi}}$ satisfy the relations (9) of the preceding paragraph. Formula (10) of § 113 gives, for the root $x = x_1 : x_2$ of the icosahedral equation, the expression

$$\eta(5\omega)x = \sum_{v_1} (-1)^{v_1} e^{\frac{\pi i}{60}\omega(6v_1+1)^2}. \quad (13)$$

The other roots are obtained by applying the icosahedral substitutions to x . But they can also be formed by replacing the variable ω by an equivalent value (Vol. I, § 120).

³Compare: Weber, "Elliptische Functionen und algebraische Zahlen". Braunschweig, 1891, pp. 63, 86, 149.

Finally, we also mention briefly the solution of the icosahedral equation by hypergeometric series. Following Gauss, we understand by

$$F(\alpha, \beta, \gamma, \xi)$$

the infinite series

$$1 + \frac{\alpha\beta}{1 \cdot \gamma}\xi + \frac{\alpha(\alpha+1)\beta(\beta+1)}{1 \cdot 2 \gamma(\gamma+1)}\xi^2 + \frac{\alpha(\alpha+1)(\alpha+2)\beta(\beta+1)(\beta+2)}{1 \cdot 2 \cdot 3 \gamma(\gamma+1)(\gamma+2)}\xi^3 + \dots$$

Putting

$$z = 1728\xi,$$

the root of the icosahedral equation

$$H^3 - zf^5 = 0$$

is then given by

$$x = \xi^{1/3} \frac{F\left(\frac{11}{60}, \frac{19}{60}, \frac{4}{3}, \xi\right)}{F\left(\frac{11}{60}, -\frac{1}{60}, \frac{1}{3}, \xi\right)}. \quad (14)$$

These series converge only so long as the absolute value of ξ is a proper fraction; for other values of ξ , however, they can be replaced by similar series.⁴

⁴H. A. Schwarz, “Ueber diejenigen Fälle, in denen die Gauss’sche Reihe $F(\alpha, \beta, \gamma, x)$ eine algebraische Function ihres vierten Elementes ist”. Crelle’s Journal, vol. 75, 1872 (collected mathematical works, vol. II). F. Klein, *Vorlesungen über das Ikosaeder*, Section I, Chap. III.